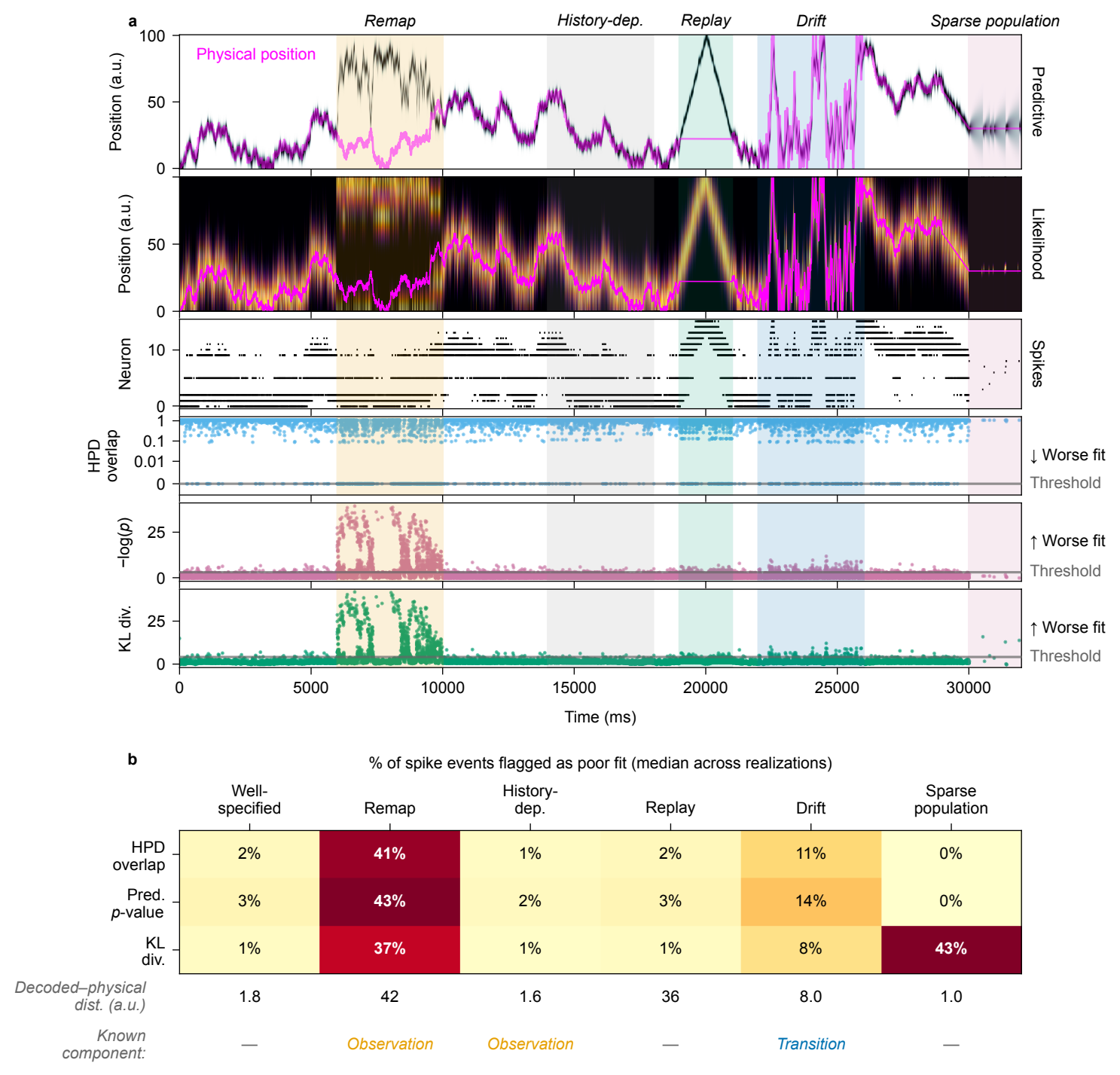

a
Remap
History-dep.
Replay
Drift
Sparse population
Physical position
Position (a.u.)
Predictive
Likelihood
Neuron
Spikes
HPD overlap
↓ Worse fit
Threshold
−log(p)
↑ Worse fit
KL div.
Time (ms)
b
% of spike events flagged as poor fit (median across realizations)
Well-specified
Remap
History-dep.
Replay
Drift
Sparse population
HPD overlap
Pred. p-value
KL div.
2% 41% 1% 2% 11% 0%
3% 43% 2% 3% 14% 0%
1% 37% 1% 1% 8% 43%
Decoded–physical dist. (a.u.)
1.8 42 1.6 36 8.0 1.0
Known component:
— Observation Observation — Transition —